2	**(a)**		torque is the product of one of the forces and the distance between forces	M1	
			the <u>perpendicular</u> distance between the forces	A1	[2]
	(b)	**(i)**	torque = 8 × 1.5 = 12 N m	A1	[1]
		(ii)	there is a resultant torque / sum of the moments is not zero	M1	
			(the rod rotates) and is not in equilibrium	A1	[2]
	(c)	**(i)**	B × 1.2 = 2.4 × 0.45	C1	
			B = 0.9(0) N	A1	[2]
		(ii)	A = 2.4 – 0.9 = 1.5 N / moments calculation	A1	[1]
3	**(a)**	**(i)**	horizontal velocity = 15 cos 60° = 7.5 m s^{-1}	A1	[1]